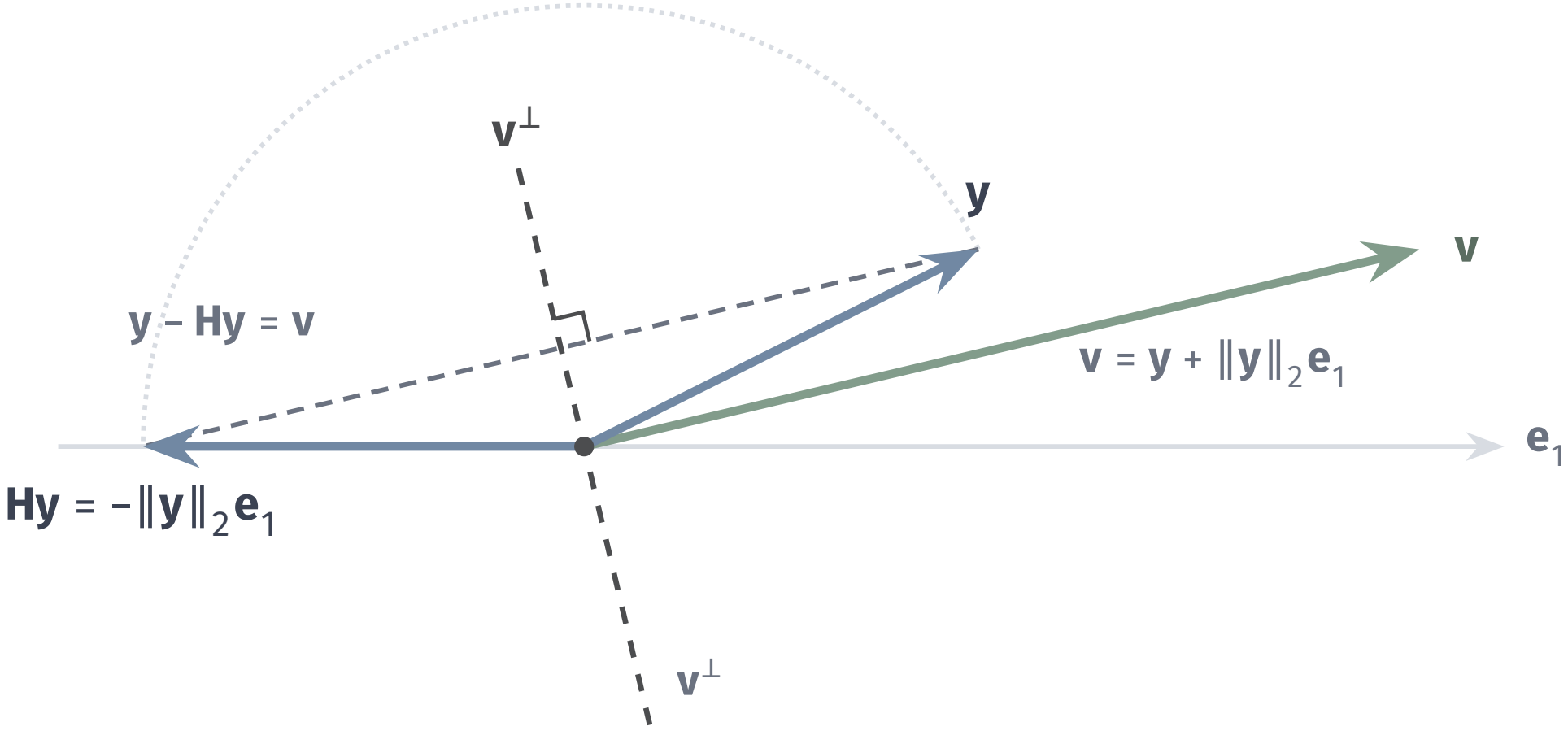
$\mathbf{v}^\perp$
$\mathbf{y}$
$\mathbf{v}$
$\mathbf{y} - \mathbf{H}\mathbf{y} = \mathbf{v}$
$\mathbf{v} = \mathbf{y} + \|\mathbf{y}\|_2 \mathbf{e}_1$
$\mathbf{e}_1$
$\mathbf{H}\mathbf{y} = -\|\mathbf{y}\|_2 \mathbf{e}_1$
$\mathbf{v}^\perp$